TEXT-BASED EMOTION DETECTION USING MACHINE LEARNING

Group Members	**Name**	**Student ID**
Member 01:	Sourav Saha	0112330638
Member 02:	Saptarshi Biswas Supty	0112330221
Member 03:	Tridib Karmokar	0112230787

Course Information:

Course Title : Artificial Intelligence Labratory
Course Code : CSE 3812
Section : I

Project Information

Project Name : Text-Based Emotion Detection Using Machine Learning

Submitted To:

Md. Tanvir Raihan

Lecturer, Department of CSE

United International University

Submission Date:

19 / 06 / 2026

1. Introduction

Emotion detection is an important application of Artificial Intelligence and Natural Language Processing (NLP). It helps computers understand human emotions from written text.

This project develops an emotion prediction system that receives user text and automatically predicts emotional categories using machine learning algorithms.

Several classification algorithms were tested and compared to identify the best performing model.

The final system also includes a Gradio interface for real-time emotion prediction.

2. Dataset Description

Dataset Collection

The dataset used in this project was collected from publicly available text-based emotion datasets and processed for multi-class emotion classification.

Dataset Source

Kaggle Dataset Repository

Dataset Link:
https://www.kaggle.com/code/shivamburnwal/speech-emotion-recognition

Search keywords: "Emotion Detection Dataset" or, "Text Emotion Classification Dataset"

Number of Samples

Approximately 100,000+ text samples.

Data Type

Structured CSV Dataset

Columns:

- Text
- Emotion Label

Emotion Classes

- Fun

- Surprise
- Neutral
- Enthusiasm
- Happiness
- Hate
- Sadness
- Empty
- Love
- Relief
- Anger

3. Data Preprocessing

The following preprocessing techniques were applied:

• Converted all text to lowercase
• Removed URLs
• Removed special characters
• Removed unnecessary spaces

TF-IDF Vectorization was used to convert text into numeric vectors.

4. Model Selection

The following machine learning models were trained and evaluated:

Logistic Regression

Linear classification model.

AdaBoost

Boosting-based classifier.

XGBoost

Advanced gradient boosting classifier.

Naive Bayes

Probability-based classifier.

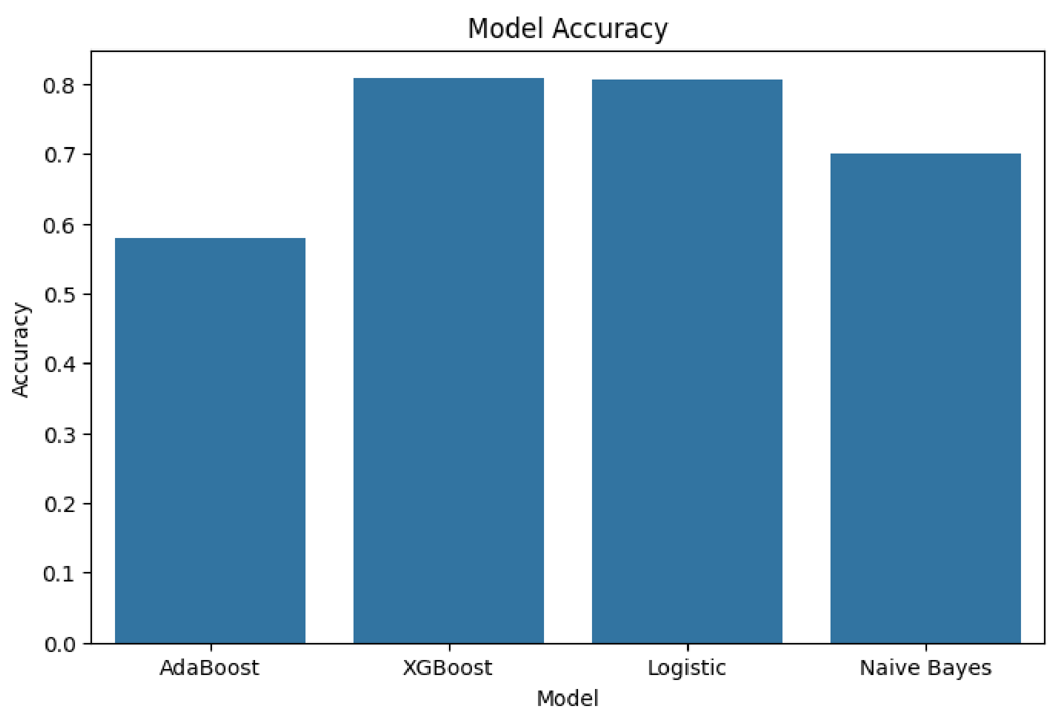
5. Experiments and Results

Accuracy Comparison

Model	Accuracy
XGBoost	80.86%
Logistic Regression	80.61%
Naive Bayes	70.02%
AdaBoost	57.84%

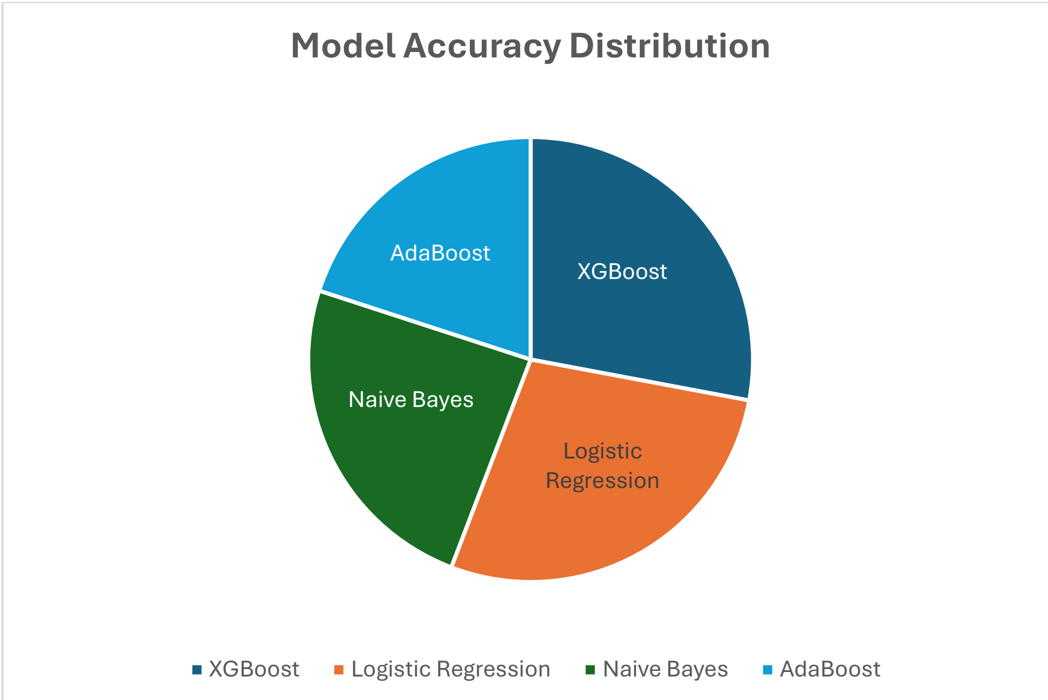
Best Model: XGBoost

Figure 1: Bar Graph



Observation: XGBoost achieved the highest accuracy.

Figure 2: Pie Chart

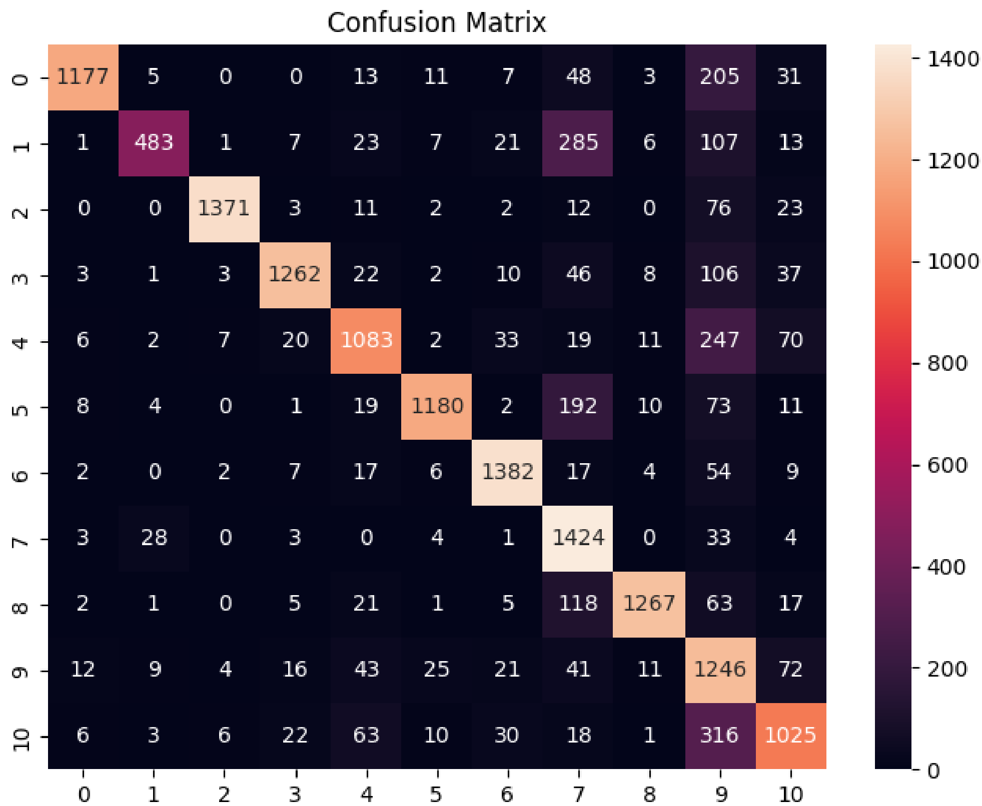


Observation:

The pie chart visually shows that:

- XGBoost and Logistic Regression contributed the highest performance.
- AdaBoost occupied the smallest accuracy portion.
- Naive Bayes achieved moderate performance.

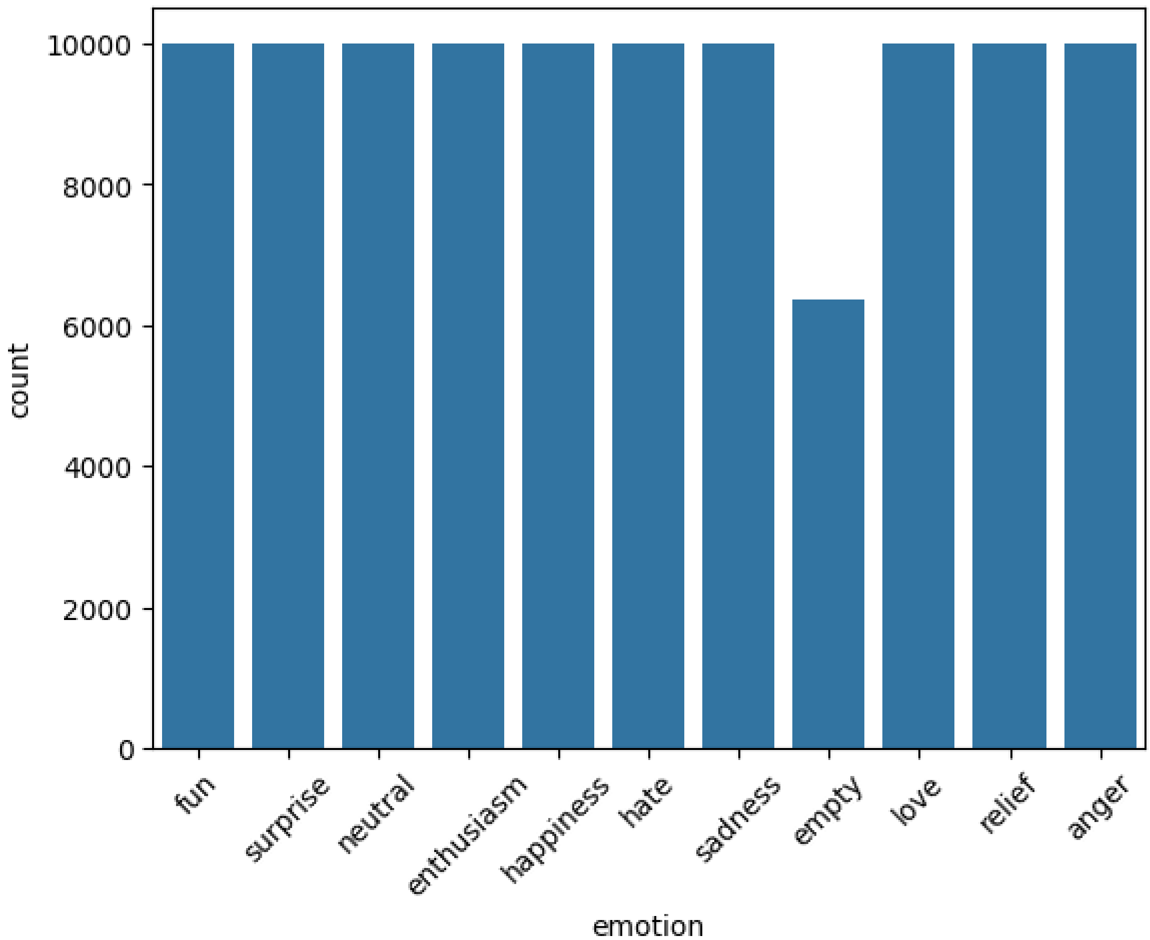
Figure 3: Confusion Matrix



Observation:

- Strong diagonal values indicate correct predictions.
- Some emotions overlap due to semantic similarity.

Figure 4: Dataset Distribution



Observation:

The dataset is mostly balanced across emotion classes.

6. Result Analysis

Experimental analysis shows that XGBoost achieved the best performance with 80.86% accuracy.

Logistic Regression also produced highly competitive results.

Naive Bayes achieved acceptable performance but was lower than XGBoost.

AdaBoost produced the weakest result among all models.

Confusion matrix analysis indicates that classes with similar emotional meaning sometimes generated classification overlap.

Balanced class distribution helped reduce model bias.

7. Conclusion

This project successfully developed a machine learning-based emotion detection system capable of identifying user emotions from text.

Among all evaluated algorithms, XGBoost produced the best classification performance.

Future improvements may include:

- Deep Learning
- BERT Models
- Real-time deployment
- Multilingual support

Application Interface:
Gradio-based prediction interface.

8. Deployment

After successful model training and evaluation, the emotion detection system was deployed as a simple interactive web application using **Gradio**.

The deployment allows users to enter text directly into the interface and instantly receive the predicted emotion.

The application loads the trained machine learning model together with the TF-IDF vectorizer and label encoder to process user input and generate predictions in real time.

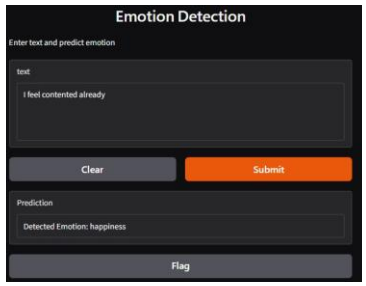
Technologies Used for Deployment

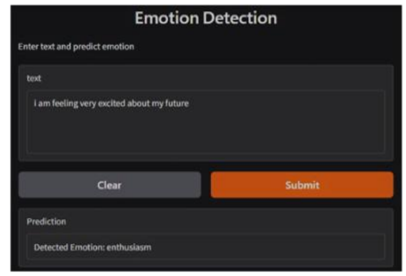
- Python
- Gradio
- Pickle
- Machine Learning Model
- TF-IDF Vectorizer

Deployment Workflow

User Input → Text Cleaning → TF-IDF Transformation → Model Prediction → Emotion Output

Deployment Screenshots





Deployment Outcome

The deployed system successfully performs real-time emotion classification and provides an easy-to-use interface for end users.

This deployment demonstrates how machine learning models can be integrated into practical applications and used outside the development environment.